PHYSICS 5054 QUESTION BANK - AUDIT
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Written 2026-11-13 08:46 UTC by tools/bank/breports.py.

RESULT: ALL AUTOMATED CHECKS PASSED; VISUAL REVIEW RECORDED

1. FILES
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physics_5054_past_paper_questions_with_ms.pdf	80,941,691 bytes 4,358 pages
physics_5054_question_bank_tracker.txt	1,400,671 bytes
physics_5054_question_bank_tracker.pdf	705,448 bytes
physics_5054_question_bank_exclusions.txt	49,846 bytes
physics_5054_question_bank_exclusions.pdf	23,276 bytes
physics_5054_question_bank_audit.txt	this file
physics_5054_question_bank_audit.pdf	PDF copy of this file

sha256 of the bank PDF: 1c4c44d6e6440b49377d820cf6fd5ce50276a796cd63adc5507c37e3b8ea7701
built 2026-11-13 08:35 UTC by tools/bank/bbuild.py; checked 2026-11-13 08:42 UTC by tools/bank/bverify.py

2. WHAT IS IN THE BANK
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records (one question in one subsection)	2,709 Paper 1: 1,723 Paper 2: 986
original questions	2,274 Paper 1: 1,723 Paper 2: 551
Paper 2 questions by number of subsections	1 subsection: 263, 2 subsections: 177, 3 subsections: 82, 4 subsections: 22, 5 subsections: 7
pages	4,358 front 2, subsection title pages 27, pages of records 4,329
pictures placed	6,261 distinct pictures: 4,951 (a question printed under several subsections uses the same pictures)
bookmarks	2,741 chapter and front matter 5, subsections 27, records 2,709
page orientation	portrait 3,923, landscape 435 (landscape mark schemes from 2017 on)
papers	156 question papers with their mark schemes: 78 of Paper 1, 78 of Paper 2, May/June 2004 to May/June 2026
EITHER/OR questions	5 questions, 6 records, both alternatives and both schemes shown; the marks in the header are those of one alternative:

4.2.2|w21|P2|V2|Q5	4 of 5 marks (printed total)
4.2.3|w09|P2|V1|Q11	2 of 15 marks (printed marks above the OR alternative)
4.2.4|w09|P2|V1|Q11	7 of 15 marks (printed marks above the OR alternative)
4.2.4|s10|P2|V1|Q7	3 of 5 marks (printed marks above the OR alternative)
4.2.4|s19|P2|V1|Q8	2 of 6 marks (printed total)
4.2.4|s21|P2|V2|Q8	3 of 6 marks (printed total)

pictures	16 grey levels, 180 dpi or more, shown at original size; text is black on white

3. RECONCILIATION

[PASS] R1 the records of the bank are exactly the question/subsection pairs that the verified findings give
2,709 pairs expected from the findings, 2,709 records in the bank, 2,709 in both
[PASS] R2 the entry list used for the build equals a fresh list made from the findings, in the same order
2,709 entries, order identical
[PASS] R3 questions of the 156 papers: in the bank + left out = all questions
Paper 1: 1,723 + 1,397 = 3,120; Paper 2: 551 + 257 = 808; checked for each of the 156 papers
[PASS] R4 no question is both in the bank and in the list of exclusions
0 questions in both lists
[PASS] R5 papers without a mark scheme: every question is listed and left out
10 papers, 200 + 55 = 255 questions
[PASS] R6 all questions of all Paper 1 and Paper 2 question papers in the folder
2,274 in the bank + 1,654 not on a selected subsection + 255 without a mark scheme = 4,183
[PASS] R7 pages of the bank: front pages + divider pages + pages of the records = all pages
2 + 27 + 4,329 = 4,358; the pages of consecutive records follow each other without a gap
[PASS] R8 files of the folder physics-5054: used + excluded + other kinds = all
311 + 12 + 471 = 794
[PASS] R9 consolidation: several tracker rows of one question in one subsection give one record
1,985 in-scope tracker rows of Paper 2 give 986 records of 551 questions; 1,723 Paper 1 questions give one record each
[PASS] R11 all papers used are complete in the paper-by-paper verification; Paper 1: the recorded answer equals the published key letter
78 Paper 1 papers: 3,120 of 3,120 answers equal the key, 0 differ or are open; 78 Paper 2 papers: 3,673 sub-questions matched to their scheme entries
[PASS] R10 the results of the checks belong to the files on disk
PDF hash equals the hash recorded by bverify.py; geometry hash equals the hash recorded by bverify.py and bcheck_geom.py

Questions by source	questions	in the bank	left out	reason for leaving out
Paper 1, papers with a mark scheme	3,120	1,723	1,397	not on a selected subsection
Paper 2, papers with a mark scheme	808	551	257	not on a selected subsection
Paper 1, papers without a mark scheme	200	0	200	no mark scheme (98 on a selected subsection)
Paper 2, papers without a mark scheme	55	0	55	no mark scheme (35 on a selected subsection, 59 question/subsection pairs)
all	4,183	2,274	1,909	

4. AUTOMATED CHECKS OF THE CROP GEOMETRY (tools/bank/bcheck_geom.py, against the source PDFs)
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run 2026-11-13 08:33 UTC; 156 papers
[PASS] V1 Paper 2: the mark allocations printed inside the crops of a question add up to the marks of that question
808 questions, 0 differ
[PASS] V2 question papers: every text row and every answer line lies in exactly one crop
48,856 rows, 0 outside or in two crops (509 ruled lines on the shared lined pages are not reproduced, see the exclusions file)
[PASS] V3 every question and every scheme starts inside the first crop of its own question
3,928 question starts, 0 misplaced
[PASS] V4 Paper 2 schemes: every row of the used scheme pages lies in exactly one crop, or is general material listed in the exclusions file
10,173 rows; 103 rows are in no crop: 77 after the scheme of the last question, 26 before the scheme of Question 1 (covers, general notes, legends of the mark codes; listed in the exclusions file, part D)
[PASS] V5 Paper 1: one crop per question; its text equals the question text of the verification
3,120 questions, 0 differ
[PASS] V6 Paper 1: the crop of a key row holds the question number and the key letter of that question, nothing else
3,120 key rows, 0 differ
[PASS] V7 the pixel row of every cut line is free of ink (more than 8 dark pixels at 2 pixels per point), or the place was inspected
10,628 cut lines, 2 with ink: s04_2 qp p. 11 at y=795.6; s25_22 qp p. 6 at y=796.5 (the two places described under V11)
[PASS] V8 Paper 2 schemes: no two neighbouring questions whose mark codes are off by opposite amounts (a row filed under the wrong question)
0 such pairs
[PASS] V9 every question crop holds text (no crop of ruled lines only)
0 crops without text
[PASS] V10 no ink is left outside the crops between page header and page footer
2,610 question-paper pages and 468 scheme pages; 0 pages with unexplained ink; 19 shared lined pages not reproduced; 16 scheme pages with the general material of V4
[PASS] V11 no stroke crosses a cut line (8 pixels per point; excepted: cuts placed on a table rule, the rule of the examiner column), or the place was inspected
10,628 cut lines, 2 crossed: s04_2 qp p. 11 at y=795.6: the mark '[9]', last line of Question 10, touches the page footer '[Turn over' (gap 0.12 pt, found at 8 pixels per point); the cut is at the lower end of the brackets and '[9]' is complete in the picture; s25_22 qp p. 6 at y=796.5: the corner marks of the page (page furniture, at x = 35-38 pt and 557-560 pt) stand beside the last answer line of Question 4; the cut is in the blank row below that line and passes through the corner marks only

Notes of the geometry program:
s04_2: QP page 11: last line of the question touches the page footer: bottom edge put at y=795.6, at a gap of 0.12 pt found at 8 pixels per point
s26_22: MS page 11: frame or rule ends directly above the page footer: bottom limit moved down to the blank row at y=549.75

5. AUTOMATED CHECKS OF THE FINISHED PDF (tools/bank/bverify.py: the PDF reopened from disk, every page rendered)

[PASS] A1 page count of the PDF equals the build manifest
4358 pages
[PASS] A2 every entry is in the PDF and no record ID occurs twice
2709 entries in page order, 2709 expected
[PASS] A3 order: subsection, year, session, variant, paper, question number
strictly increasing sort key
[PASS] A4 one record per original question and subsection
2274 original questions, 2709 records
[PASS] A5 pages of an entry are adjacent; question pictures first, complete scheme directly after, in source order
0 entries with a problem
[PASS] A6 bookmarks: one per chapter, subsection and entry, each pointing to the first page of its target
5 top-level, 27 subsection and 2709 question bookmarks
[PASS] A7 file size below 100 MB
80.9 MB (80941691 bytes)
[PASS] B0 every page of the PDF renders without a MuPDF error or warning
4358 pages rendered; warnings: none
[PASS] B1 no page is dark or colour-inverted (every page rendered)
0 dark pages of 4358
[PASS] B2 no blank page and no empty picture
0 findings
[PASS] B3a every picture is on its page at the planned position (picture boxes read back from the PDF)
6261 placements, 0 not found
[PASS] B3b the stored pixels of every distinct picture are identical to a fresh rendering of its region of the source PDF
4951 distinct pictures, 0 differ
[PASS] B3c every page, rendered on the pixel grid of its pictures, shows exactly the stored pictures (nothing over them, none displaced)
6261 placements compared pixel by pixel, 0 differ
[PASS] B4 nothing printed outside the pictures and the grey bars
0 pages with stray ink
[PASS] B5 pictures lie inside the page area, below the header and above the footer, without overlap; orientation as planned; no page rotation
0 findings
[PASS] B6 the cut lines (outermost pixel rows of every picture) are free of ink; rows of the key table and cuts on a table rule excepted
6453 cut lines checked, 3449 on a ruled line by design; 6 with ink, of which 6 reviewed and accepted (listed in the audit), 0 not reviewed
[PASS] B7 every picture is shown at its original size (99.9 % or more) and at 180 dpi or more
6261 placements; scales {0.999: 50, 1.0: 3577, 1.001: 1795, 1.002: 223, 1.003: 281, 1.004: 97, 1.005: 24, 1.006: 214}; lowest resolution 180 dpi

[PASS] C1 header, source lines, record ID and page number of every page agree with the entry
0 pages with a missing text
[PASS] C2 marks of the question named in the header of every Paper 2 record = marks printed on the question paper inside its pictures
986 records of 551 questions: 980 records without alternatives (sum of all printed '[n]' and the printed '[Total: n]'), 6 records of EITHER/OR questions (marks of one alternative); 0 differ

Cut lines with ink that were inspected by eye and accepted (B6):

- bank page 238, bottom edge, 5054_s05_qp_2.pdf source page 2: the mark '[1]' of the last answer line ends directly above the page footer; the brackets are complete and nothing crosses the cut line
- bank page 720, bottom edge, 5054_s04_qp_2.pdf source page 11: the mark '[9]', last line of Question 10, touches the footer '[Turn over' (gap 0.12 pt); the cut is at the lower end of the brackets, '[9]' is complete; the last pixel row may carry a faint grey trace of the top of the footer text
- bank page 2782, bottom edge, 5054_s25_qp_22.pdf source page 6: the corner marks of the page (page furniture) stand beside the last answer line of Question 4; the cut is in the blank row below that line and passes through the two corner marks only; nothing of the question is cut
- bank page 3052, bottom edge, 5054_s26_ms_22.pdf source page 11: the frame of the note 'bod' below the table ends 0.5 pt above the page footer; the frame is complete
- bank page 4315, bottom edge, 5054_s23_qp_21.pdf source page 5: '[Total: 6]' ends directly above the footer '[Turn over'; the brackets are complete and nothing crosses the cut line
- bank page 4318, bottom edge, 5054_s23_qp_22.pdf source page 11: '[Total: 9]' ends directly above the footer '[Turn over'; the brackets are complete and nothing crosses the cut line

6. RECORDS PER SUBSECTION

subsection		Paper 1	Paper 2	records	bank pages
1.1	Physical quantities and measurement techniques	156	26	182	3-229
1.2	Motion	100	76	176	230-537
1.3	Mass and weight	73	39	112	538-717
1.4	Density	57	35	92	718-867
1.5.1	Balanced and unbalanced forces	85	62	147	868-1118
1.5.2	Friction	66	45	111	1119-1309
1.5.3	Elastic deformation	62	22	84	1310-1432
1.5.4	Circular motion	52	15	67	1433-1525
1.5.5	Turning effect of forces	59	37	96	1526-1679
1.5.6	Centre of gravity	41	14	55	1680-1759
1.6	Momentum	8	11	19	1760-1798
1.7.1	Energy	68	83	151	1799-2085
1.7.2	Work	38	30	68	2086-2199
1.7.3	Energy resources	27	17	44	2200-2272
1.7.4	Efficiency	41	29	70	2273-2389
1.7.5	Power	60	48	108	2390-2573
3.1	General properties of waves	89	53	142	2574-2798
3.2.1	Reflection of light	55	9	64	2799-2877
3.2.2	Refraction of light	76	37	113	2878-3052
3.2.3	Thin lenses	71	22	93	3053-3179
3.2.4	Dispersion of light	29	13	42	3180-3243
3.3	Electromagnetic spectrum	53	46	99	3244-3410
3.4	Sound	115	38	153	3411-3623
4.2.1	Electrical charge	90	29	119	3624-3783
4.2.2	Electrical current	35	40	75	3784-3932
4.2.3	Electromotive force and potential difference	44	29	73	3933-4062
4.2.4	Resistance	73	81	154	4063-4358
all		1723	986	2709	

7. RECORDS PER PAPER

paper	question paper	mark scheme	questions	in the bank	records	left out
s04_1	5054_s04_qp_1.pdf	5054_s04_ms.pdf	40	20	20	20
s04_2	5054_s04_qp_2.pdf	5054_s04_ms.pdf	11	6	11	5
w04_1	5054_w04_qp_1.pdf	5054_w04_ms_1.pdf	40	20	20	20
w04_2	5054_w04_qp_2.pdf	5054_w04_ms_2.pdf	10	7	8	3
s05_1	5054_s05_qp_1.pdf	5054_s05_ms_1.pdf	40	18	18	22
s05_2	5054_s05_qp_2.pdf	5054_s05_ms_2.pdf	11	7	14	4
w05_1	5054_w05_qp_1.pdf	5054_w05_ms_1.pdf	40	21	21	19
w05_2	5054_w05_qp_2.pdf	5054_w05_ms_2.pdf	11	9	16	2
s06_1	5054_s06_qp_1.pdf	5054_s06_ms_1.pdf	40	21	21	19
s06_2	5054_s06_qp_2.pdf	5054_s06_ms_2.pdf	11	7	9	4
w06_1	5054_w06_qp_1.pdf	5054_w06_ms_1.pdf	40	17	17	23
w06_2	5054_w06_qp_2.pdf	5054_w06_ms_2.pdf	11	8	12	3
s07_1	5054_s07_qp_1.pdf	5054_s07_ms_1.pdf	40	22	22	18
s07_2	5054_s07_qp_2.pdf	5054_s07_ms_2.pdf	11	8	13	3
w07_1	5054_w07_qp_1.pdf	5054_w07_ms_1.pdf	40	20	20	20
w07_2	5054_w07_qp_2.pdf	5054_w07_ms_2.pdf	11	7	11	4
s08_1	5054_s08_qp_1.pdf	5054_s08_ms_1.pdf	40	20	20	20
s08_2	5054_s08_qp_2.pdf	5054_s08_ms_2.pdf	11	7	11	4
w08_1	5054_w08_qp_1.pdf	5054_w08_ms_1.pdf	40	20	20	20
w08_2	5054_w08_qp_2.pdf	5054_w08_ms_2.pdf	11	8	15	3
s09_1	5054_s09_qp_1.pdf	5054_s09_ms_1.pdf	40	18	18	22
s09_2	5054_s09_qp_2.pdf	5054_s09_ms_2.pdf	11	6	11	5
w09_1	5054_w09_qp_1.pdf	5054_w09_ms_1.pdf	40	20	20	20
w09_2	5054_w09_qp_2.pdf	5054_w09_ms_2.pdf	11	8	13	3
s10_11	5054_s10_qp_11.pdf	5054_s10_ms_11.pdf	40	23	23	17
s10_12	5054_s10_qp_12.pdf	5054_s10_ms_12.pdf	40	23	23	17

s10_21	5054_s10_qp_21.pdf	5054_s10_ms_21.pdf	11	9	14	2
s10_22	5054_s10_qp_22.pdf	5054_s10_ms_22.pdf	11	8	14	3
w10_11	5054_w10_qp_11.pdf	5054_w10_ms_11.pdf	40	20	20	20
w10_12	5054_w10_qp_12.pdf	5054_w10_ms_12.pdf	40	20	20	20
w10_21	5054_w10_qp_21.pdf	5054_w10_ms_21.pdf	10	9	14	1
w10_22	5054_w10_qp_22.pdf	5054_w10_ms_22.pdf	11	8	16	3
s11_11	5054_s11_qp_11.pdf	5054_s11_ms_11.pdf	40	22	22	18
s11_12	5054_s11_qp_12.pdf	5054_s11_ms_12.pdf	40	22	22	18
s11_21	5054_s11_qp_21.pdf	5054_s11_ms_21.pdf	11	8	11	3
s11_22	5054_s11_qp_22.pdf	5054_s11_ms_22.pdf	11	8	12	3
w11_11	5054_w11_qp_11.pdf	5054_w11_ms_11.pdf	40	21	21	19
w11_12	5054_w11_qp_12.pdf	5054_w11_ms_12.pdf	40	21	21	19
w11_21	5054_w11_qp_21.pdf	5054_w11_ms_21.pdf	11	8	14	3
w11_22	5054_w11_qp_22.pdf	5054_w11_ms_22.pdf	11	8	15	3
s12_11	5054_s12_qp_11.pdf	5054_s12_ms_11.pdf	40	21	21	19
s12_12	5054_s12_qp_12.pdf	5054_s12_ms_12.pdf	40	21	21	19
s12_21	5054_s12_qp_21.pdf	5054_s12_ms_21.pdf	11	9	13	2
s12_22	5054_s12_qp_22.pdf	5054_s12_ms_22.pdf	10	7	11	3
w12_11	5054_w12_qp_11.pdf	5054_w12_ms_11.pdf	40	20	20	20
w12_12	5054_w12_qp_12.pdf	5054_w12_ms_12.pdf	40	20	20	20
w12_21	5054_w12_qp_21.pdf	5054_w12_ms_21.pdf	11	10	16	1
w12_22	5054_w12_qp_22.pdf	5054_w12_ms_22.pdf	11	8	12	3
s13_11	5054_s13_qp_11.pdf	5054_s13_ms_11.pdf	40	24	24	16
s13_12	5054_s13_qp_12.pdf	5054_s13_ms_12.pdf	40	24	24	16
s13_21	5054_s13_qp_21.pdf	5054_s13_ms_21.pdf	11	5	8	6
s13_22	5054_s13_qp_22.pdf	5054_s13_ms_22.pdf	11	9	17	2
w13_11	5054_w13_qp_11.pdf	5054_w13_ms_11.pdf	40	21	21	19
w13_12	5054_w13_qp_12.pdf	5054_w13_ms_12.pdf	40	19	19	21
w13_21	5054_w13_qp_21.pdf	5054_w13_ms_21.pdf	11	6	13	5
w13_22	5054_w13_qp_22.pdf	5054_w13_ms_22.pdf	11	8	18	3
s14_11	5054_s14_qp_11.pdf	5054_s14_ms_11.pdf	40	23	23	17
s14_12	5054_s14_qp_12.pdf	5054_s14_ms_12.pdf	40	21	21	19
s14_21	5054_s14_qp_21.pdf	5054_s14_ms_21.pdf	11	8	15	3
s14_22	5054_s14_qp_22.pdf	5054_s14_ms_22.pdf	11	8	13	3
w14_11	5054_w14_qp_11.pdf	5054_w14_ms_11.pdf	40	19	19	21
w14_12	5054_w14_qp_12.pdf	5054_w14_ms_12.pdf	40	18	18	22
w14_21	5054_w14_qp_21.pdf	5054_w14_ms_21.pdf	11	7	16	4
w14_22	5054_w14_qp_22.pdf	5054_w14_ms_22.pdf	11	7	12	4
s15_11	5054_s15_qp_11.pdf	5054_s15_ms_11.pdf	40	23	23	17
s15_12	5054_s15_qp_12.pdf	5054_s15_ms_12.pdf	40	20	20	20
s15_21	5054_s15_qp_21.pdf	5054_s15_ms_21.pdf	11	7	12	4
s15_22	5054_s15_qp_22.pdf	5054_s15_ms_22.pdf	11	7	14	4
w15_11	5054_w15_qp_11.pdf	5054_w15_ms_11.pdf	40	20	20	20
w15_12	5054_w15_qp_12.pdf	5054_w15_ms_12.pdf	40	24	24	16
w15_21	5054_w15_qp_21.pdf	5054_w15_ms_21.pdf	11	8	15	3
w15_22	5054_w15_qp_22.pdf	5054_w15_ms_22.pdf	11	7	13	4
s16_11	5054_s16_qp_11.pdf	5054_s16_ms_11.pdf	40	27	27	13
s16_12	5054_s16_qp_12.pdf	5054_s16_ms_12.pdf	40	20	20	20
s16_21	5054_s16_qp_21.pdf	5054_s16_ms_21.pdf	12	9	15	3
s16_22	5054_s16_qp_22.pdf	5054_s16_ms_22.pdf	11	8	11	3
w16_11	5054_w16_qp_11.pdf	5054_w16_ms_11.pdf	40	21	21	19
w16_12	5054_w16_qp_12.pdf	5054_w16_ms_12.pdf	40	24	24	16
w16_21	5054_w16_qp_21.pdf	5054_w16_ms_21.pdf	10	7	13	3
w16_22	5054_w16_qp_22.pdf	5054_w16_ms_22.pdf	10	6	15	4
s17_11	5054_s17_qp_11.pdf	5054_s17_ms_11.pdf	40	26	26	14
s17_12	5054_s17_qp_12.pdf	5054_s17_ms_12.pdf	40	23	23	17
s17_21	5054_s17_qp_21.pdf	5054_s17_ms_21.pdf	11	7	13	4
s17_22	5054_s17_qp_22.pdf	5054_s17_ms_22.pdf	11	8	13	3
w17_11	5054_w17_qp_11.pdf	5054_w17_ms_11.pdf	40	22	22	18
w17_12	5054_w17_qp_12.pdf	5054_w17_ms_12.pdf	40	21	21	19
w17_21	5054_w17_qp_21.pdf	5054_w17_ms_21.pdf	11	5	14	6
w17_22	5054_w17_qp_22.pdf	5054_w17_ms_22.pdf	11	10	20	1
s18_11	5054_s18_qp_11.pdf	5054_s18_ms_11.pdf	40	20	20	20
s18_12	5054_s18_qp_12.pdf	5054_s18_ms_12.pdf	40	23	23	17
s18_21	5054_s18_qp_21.pdf	5054_s18_ms_21.pdf	11	8	13	3
s18_22	5054_s18_qp_22.pdf	5054_s18_ms_22.pdf	11	7	14	4
w18_11	5054_w18_qp_11.pdf	5054_w18_ms_11.pdf	40	23	23	17
w18_12	5054_w18_qp_12.pdf	5054_w18_ms_12.pdf	40	20	20	20
w18_21	5054_w18_qp_21.pdf	5054_w18_ms_21.pdf	10	7	15	3
w18_22	5054_w18_qp_22.pdf	5054_w18_ms_22.pdf	10	8	16	2
s19_11	5054_s19_qp_11.pdf	5054_s19_ms_11.pdf	40	22	22	18
s19_12	5054_s19_qp_12.pdf	5054_s19_ms_12.pdf	40	28	28	12
s19_21	5054_s19_qp_21.pdf	5054_s19_ms_21.pdf	11	8	16	3
s19_22	5054_s19_qp_22.pdf	5054_s19_ms_22.pdf	11	7	11	4
w19_11	5054_w19_qp_11.pdf	5054_w19_ms_11.pdf	40	21	21	19
w19_12	5054_w19_qp_12.pdf	5054_w19_ms_12.pdf	40	22	22	18
w19_21	5054_w19_qp_21.pdf	5054_w19_ms_21.pdf	10	5	10	5
w19_22	5054_w19_qp_22.pdf	5054_w19_ms_22.pdf	10	7	13	3
s20_11	5054_s20_qp_11.pdf	5054_s20_ms_11.pdf	40	25	25	15
s20_12	5054_s20_qp_12.pdf	5054_s20_ms_12.pdf	40	27	27	13
s20_21	5054_s20_qp_21.pdf	5054_s20_ms_21.pdf	10	7	11	3
s20_22	5054_s20_qp_22.pdf	5054_s20_ms_22.pdf	10	7	12	3
w20_11	5054_w20_qp_11.pdf	5054_w20_ms_11.pdf	40	26	26	14

w20_12	5054_w20_qp_12.pdf	5054_w20_ms_12.pdf	40	24	24	16
w20_21	5054_w20_qp_21.pdf	5054_w20_ms_21.pdf	10	7	11	3
w20_22	5054_w20_qp_22.pdf	5054_w20_ms_22.pdf	9	5	12	4
s21_11	5054_s21_qp_11.pdf	5054_s21_ms_11.pdf	40	22	22	18
s21_12	5054_s21_qp_12.pdf	5054_s21_ms_12.pdf	40	24	24	16
s21_21	5054_s21_qp_21.pdf	5054_s21_ms_21.pdf	10	6	13	4
s21_22	5054_s21_qp_22.pdf	5054_s21_ms_22.pdf	11	9	15	2
w21_11	5054_w21_qp_11.pdf	5054_w21_ms_11.pdf	40	24	24	16
w21_12	5054_w21_qp_12.pdf	5054_w21_ms_12.pdf	40	23	23	17
w21_21	5054_w21_qp_21.pdf	5054_w21_ms_21.pdf	10	7	13	3
w21_22	5054_w21_qp_22.pdf	5054_w21_ms_22.pdf	9	7	14	2
s22_11	5054_s22_qp_11.pdf	5054_s22_ms_11.pdf	40	25	25	15
s22_12	5054_s22_qp_12.pdf	5054_s22_ms_12.pdf	40	25	25	15
s22_21	5054_s22_qp_21.pdf	5054_s22_ms_21.pdf	9	6	13	3
s22_22	5054_s22_qp_22.pdf	5054_s22_ms_22.pdf	9	6	12	3
w22_11	5054_w22_qp_11.pdf	5054_w22_ms_11.pdf	40	21	21	19
w22_12	5054_w22_qp_12.pdf	5054_w22_ms_12.pdf	40	27	27	13
w22_21	5054_w22_qp_21.pdf	5054_w22_ms_21.pdf	9	7	13	2
w22_22	5054_w22_qp_22.pdf	5054_w22_ms_22.pdf	9	6	9	3
s23_11	5054_s23_qp_11.pdf	5054_s23_ms_11.pdf	40	21	21	19
s23_12	5054_s23_qp_12.pdf	5054_s23_ms_12.pdf	40	21	21	19
s23_21	5054_s23_qp_21.pdf	5054_s23_ms_21.pdf	11	6	8	5
s23_22	5054_s23_qp_22.pdf	5054_s23_ms_22.pdf	10	5	10	5
w23_11	5054_w23_qp_11.pdf	5054_w23_ms_11.pdf	40	19	19	21
w23_12	5054_w23_qp_12.pdf	5054_w23_ms_12.pdf	40	24	24	16
w23_21	5054_w23_qp_21.pdf	5054_w23_ms_21.pdf	9	6	12	3
w23_22	5054_w23_qp_22.pdf	5054_w23_ms_22.pdf	10	7	11	3
s24_11	5054_s24_qp_11.pdf	5054_s24_ms_11.pdf	40	21	21	19
s24_12	5054_s24_qp_12.pdf	5054_s24_ms_12.pdf	40	25	25	15
s24_21	5054_s24_qp_21.pdf	5054_s24_ms_21.pdf	8	6	10	2
s24_22	5054_s24_qp_22.pdf	5054_s24_ms_22.pdf	9	6	10	3
w24_11	5054_w24_qp_11.pdf	5054_w24_ms_11.pdf	40	26	26	14
w24_12	5054_w24_qp_12.pdf	5054_w24_ms_12.pdf	40	22	22	18
w24_21	5054_w24_qp_21.pdf	5054_w24_ms_21.pdf	9	5	10	4
w24_22	5054_w24_qp_22.pdf	5054_w24_ms_22.pdf	9	4	7	5
s25_11	5054_s25_qp_11.pdf	5054_s25_ms_11.pdf	40	20	20	20
s25_12	5054_s25_qp_12.pdf	5054_s25_ms_12.pdf	40	22	22	18
s25_21	5054_s25_qp_21.pdf	5054_s25_ms_21.pdf	9	5	12	4
s25_22	5054_s25_qp_22.pdf	5054_s25_ms_22.pdf	9	6	11	3
w25_11	5054_w25_qp_11.pdf	5054_w25_ms_11.pdf	40	24	24	16
w25_12	5054_w25_qp_12.pdf	5054_w25_ms_12.pdf	40	25	25	15
w25_21	5054_w25_qp_21.pdf	5054_w25_ms_21.pdf	9	6	10	3
w25_22	5054_w25_qp_22.pdf	5054_w25_ms_22.pdf	9	5	12	4
s26_11	5054_s26_qp_11.pdf	5054_s26_ms_11.pdf	40	26	26	14
s26_12	5054_s26_qp_12.pdf	5054_s26_ms_12.pdf	40	24	24	16
s26_21	5054_s26_qp_21.pdf	5054_s26_ms_21.pdf	9	7	10	2
s26_22	5054_s26_qp_22.pdf	5054_s26_ms_22.pdf	9	5	11	4
all			3928	2274	2709	1654

8. VISUAL REVIEW

Record of the inspection by eye, written by the AI agent that built the bank. It describes the file physics_5054_past_paper_questions_with_ms.pdf with sha256 1c4c44d6e6440b49 (first 16 digits), 4,358 pages. "Viewed" means: the page was rendered from the PDF on disk and the image was looked at.

8.1 What was viewed

* Contact sheets of ALL 4,358 pages of the file described here (41 sheets of small page images written by bverify.py), read at half size: every page is white with black print; no page is dark or colour-inverted; no page is blank; the landscape mark schemes stand upright; the front pages and the 27 title pages are in place.
* The same 41 sheets had been read at full size for an earlier build (12 November 2026, 19:06 UTC). The file described here was compared with that build page by page (both rendered at 72 dpi; pixels and page text compared).
  70 pages differ: the front page, the header bars of the 62 pages of the 25 records corrected in 8.3, and the scheme pictures of the 8 pages of the last correction in 8.3 (page 3246 is in both groups). The other 4,288 pages are identical, pixel for pixel. Viewed again after the change: the front page; the header bars of 1366, 2880, 2928, 3905, 4110, 4252, 4285 and 4293; all 8 pages with a changed scheme picture.
* 83 pages at full size (of 20 of them the upper part only):
  front page, contents, first title page (3): 1-3
  Paper 1, question with its key row (12): 4, 7, 159, 229, 642, 701, 714, 1824, 2478, 3462, 3503, 3765
  Paper 2, question pages (45): 238, 390, 398, 409, 457, 465, 546, 720, 870, 1015, 1366, 1511, 1897, 2073, 2744-2746, 2777, 2880-2881, 2908, 2928, 3150, 3293, 3296, 3360, 3372, 3761-3762, 3779-3780, 3848, 3892-3893, 3905-3906, 4102-4103, 4110, 4115, 4149, 4252-4253, 4293-4294
  Paper 2, mark schemes on portrait pages (13): 307, 437, 444, 721, 771, 871, 2882, 3246, 3628, 3894, 4104, 4111, 4150
  Paper 2, mark schemes on landscape pages (10): from source pages that carry a page rotation of 90 degrees: 127, 1963, 2747-2748, 2845, 3710, 3907; from landscape source pages without rotation: 532, 3052, 3153
  and 4 pages at 62 % size: 1893, 2117, 2476, 2969 (the scheme pictures of 771 and 2845 under other subsections).
  Chosen for: the first record; the five EITHER/OR questions (3905-3907, 4102-4104, 4110-4111, 4252-4253, 4293-4294); a record of five pages, the longest there is (2744-2748); the oldest papers (2004: 720-721, 870-871, 2880-2882, 3246, 3628) and the newest (2026: 229, 532, 3052, with the pale watermark of the download site); schemes in small print (437, 444); long header lines (3150, 3761-3762); hatching and dot patterns (2073, 2777, 3779-3780); every page that looks dark or almost empty on the contact sheets; a random sample of 16 pages (seed

5054: 307, 409, 465, 642, 714, 1511, 1824, 1897, 1963, 2478, 3153, 3372, 3503, 3628, 3765, 3848).
On all of them the question or scheme is complete, upright and black on white, the header names the right paper and question, and the scheme follows its question. On the Paper 1 pages the key letter stands under the question.
* The six cut lines that carry ink (check B6; listed in part 5), each enlarged beside its source page.
* The two cut lines of the geometry check that a stroke touches (V7, V11; described in part 4), enlarged.

8.2 Observations that are not defects
* Dark areas on the contact sheets are artwork of the papers: 398, 1015 and 1483 (photograph of an astronaut, 5054_w17_qp_22 p. 13), 457, 677 and 1065 (man in a lift, 5054_w21_qp_21 p. 4), 546 (5054_s06_qp_2 p. 11), 701 (5054_s24_qp_12 p. 3), 3293 (5054_s13_qp_22 p. 6), 3296 (5054_w13_qp_21 p. 5), 3360 and 3548 (X-ray picture, 5054_s20_qp_21 p. 17).
* Pages that look almost empty are genuine: 2908, 4115 and 4149 hold drawing or answer space; 2881 holds a picture 38 pt high with the single mark "[2]" (beside it the page number and the examiner label of the paper), because the paper prints the last line of that question at the top of its next page.
* Page furniture inside pictures (described with counts in the exclusions file, part D2): the heading of a section above its first question; the page number in the Paper 2 papers of 2004-2007 (seen on 238, 2880, 2881); a notice "Question 9 begins over the page." after the last line of a question (seen on 2746); the column "For Examiner's Use"; the vertical notice "DO NOT WRITE IN THIS MARGIN" of the papers of 2025 and 2026, which is cut where a picture ends (seen on 2073, 2777, 3779, 3780).

8.3 Defects found by this inspection and corrected in the file described here
* Marks of EITHER/OR questions. The header of 4110-4111 (May/June 2010, Paper 2 variant 1, Question 7) said "3 of 7 marks" and that of 4293-4295 (May/June 2021, variant 2, Question 8) "3 of 9 marks": the marks of both alternatives had been added. A candidate answers one alternative, so the questions carry 5 and 6 marks. These two questions and May/June 2019 variant 1 Question 8 (4252-4254) also lacked the remark "EITHER / OR question" in the header. Cause: alternatives were recognised only where the tracker has separate rows for the OR part. They are now recognised from the text of the question paper. Check C2 of bverify.py was written for this: it compares the marks in the header of every Paper 2 record with the marks printed on the paper (part 5).
* Names of the parts in the header. Where the paper prints one mark for several sub-parts, e.g. one "[2]" after (a)(i) and (a)(ii), the header named the last sub-part only, "(a)(ii)". 21 records, mostly of the two papers of 2004 (seen on 2882: "(a)(ii), (b), (c)(ii) (7 of 7 marks)" for a scheme that also marks (a)(i) and (c)(i)). The header now names every sub-part the mark covers: "(a)(i)-(ii), (b), (c)(i)-(ii)".
* Record 4.2.2|w21|P2|V2|Q5 (3905-3907): the whole question is "5 EITHER ... OR ..."; the header now says "EITHER (a), EITHER (b)(ii)" instead of "(a), (b)(ii)".
* The front page and the exclusions file said that page numbers and section headings are never inside a picture. That was not accurate (see 8.2); both texts were corrected and the exclusions file gives the counts.
* A heading shown with the wrong scheme. The mark scheme of May/June 2017 variant 2 prints the heading "SECTION B" in capitals at the top of the page that follows the scheme of Question 8, the last question of Section A. The heading was shown as a strip of its own below the scheme of Question 8 (3710). Two more schemes print it in capitals (May/June 2004 and May/June 2014 variant 1); their Question 8 is not in the bank. The heading is now kept with the first question of Section B, as in all other schemes: 3710 shows the scheme of Question 8 only, and the scheme of Question 9 of the three papers begins with the heading (771, 1893, 2117, 2476, 2845, 2969, 3246). No other picture of a scheme holds a heading that is not its first line (exclusions file, part D2).

Defects of earlier builds, found by the automated checks, viewed before and after the correction:
* three pictures with hatching or dot patterns (2073, 2777, 3779) were not identical to a fresh rendering of the source (check B3b); the renderer reused a bitmap decoded at another size. All pictures were rendered again with the renderer's store emptied before each page;
* the frame of a note below a table of the scheme of May/June 2026 variant 2 Question 5 (3052) and the last line "[9]" of May/June 2004 Question 10 (720, 2391) reach down to the page footer and were cut by the lower edge of the picture; the edge was moved and both are complete (part 4, notes of the geometry program);
* long header lines were shortened with "..."; the type is now reduced instead and nothing is left out (3150, 3761-3762). No header of the file contains "...", and every page carries the full title of its subsection.

8.4 Not inspected by eye
* 4,271 of the 4,358 pages were seen only as small images on the contact sheets. At that size one can see that a page is complete, upright and black on white, but not read it. That every picture shows exactly its region of the source page, and that every line of every question and scheme lies in a picture, rests on the automated checks of parts 4 and 5.
* Whether each question is filed under the best subsection was not reviewed again in this inspection.

9. HOW THE BANK WAS MADE, AND LIMITS
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* Source: the PDF files in the folder physics-5054 of the repository. Every picture of the bank is rendered directly from them
by tools/bank/bbuild.py. No page of an earlier bank is used.
* The files of the repository are stored with inverted colours: each page content is wrapped in a block that paints the page
white and inverts it with a blend mode. The wrapper also leaves white as the fill colour, so text that the original draws in
the default colour is invisible in the stored copy (282 pages of 31 of the files used, mostly papers of 2004-2009).
The bank removes the wrapper from the page content before rendering; it does not invert pixels. All pages of the bank are
black on white (check B1) and the text that is invisible in the stored copies is present.
* Landscape mark schemes (2017 on) carry a page rotation or are landscape pages. The rotation is removed with the display
transformation of the page before any coordinate is measured; these schemes are shown on landscape pages, upright.
* A question is cut out over the full page width from the blank band above its first line to the blank band below its last
line; a question that runs over several source pages gives one picture per page. The cut between two questions is placed
from the ink of the rendered page, not from text recognition. Nothing inside a picture is moved, removed or compressed.
* Which subsection a question belongs to comes from the paper-by-paper verification of this project (every question

read with
its mark scheme by the AI agent that built the bank, plus mechanical checks). It is a judgement, not an official Cambridge classification. Questions written for earlier syllabuses are filed under the 2026-2028 subsection they fit.

* The automated checks compare pixels, positions, page text and counts. They cannot judge whether a question is filed under
the best subsection. The visual review (part 8) covered contact sheets of all pages and full-size samples, not every page
at full size.
* Third-party marks in the source files (a watermark and footer of the download site in the 2026 files) are part of the source
pages; the footer is outside the pictures, the pale watermark is visible inside them.